Proceedings of the Design Society, Volume 5: ICED25
https://doi.org/10.1017/pds.2025.10155

Early engagement in co-designing mixed reality assembly instructions in manufacturing industry

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ABSTRACT: This paper presents a case study within a small manufacturing company, engaging in the early phases of co-designing Mixed Reality assembly instructions. Using generative tools through two co-design workshops, we engaged the participants in reflecting, visualising and defining their processes, needs, challenges and future ways of working in relation to new technological applications. We gained insights into the participants' current practices and identified areas where new technologies could improve these practices. We co-designed a lock assembly instruction paper prototype to use as support for future MR development, focused on their apprenticeship training. We also uncovered other areas of technological implementation, setting the framework for co-designing a customised production system.

KEYWORDS: collaborative design, early design phases, technology, tangible design methods, mixed reality

1. Introduction

This paper focuses on co-designing a human-centred Mixed Reality assembly training instruction for lock products, in a collaborative process with a small manufacturing company, supported by a technology partner in Sweden. The aim is to explore the human-centred approach in up-skilling the manufacturing company in implementing new technologies, engaging the people in co-designing these new technologies through a series of activities.

Human-centricity is one of the core objectives with Industry 5.0, the concept introduced by the EU commission (Breque et al., 2021). A human-centred approach in industry requires that technology is developed to enable people rather than to replace them (Frey, 2018). The human-centred approach needs to be integrated by understanding people's conditions involved in industrial work activities, in relation to their organisational structures and technological applications (Rösiö et al., 2021), as well as how people can augment machines, and how machines can enhance people's full potential (Leng et al., 2022). However, this approach should not only apply by training people in how to use new technologies (Adel, 2022). Working in different industrial contexts, people have particular values, needs, and requirements as an independent group, thus these technologies need to be personalised and developed accordingly, to suit people's specific applications (Leng et al., 2022). Placing people's well-being at the centre of manufacturing systems implies engaging them in co-designing the development and application of new technologies.

One of the emerging technologies is Mixed Reality (MR), a hybrid application combining augmented and virtual reality to provide interactive experience between real and virtual settings. In the manufacturing industry, one MR tool is Microsoft HoloLens 2, a head-mounted holographic computer which aims to minimise the complexity and programming knowledge required in developing MR applications by proposing a more accessible platform for creating MR experiences (Hillman, 2023). However, despite its evolving accessibility, current solutions are still experimental, with low acceptance for daily applications (Moser, 2019). Developing MR applications requires collaborative approaches

https://doi.org/10.1017/pds.2025.10155 Published online by Cambridge University Press

between different stakeholders to ensure the usability and human centricity of the final design (Jin & Fagan, 2024).

By engaging the participants in a series of collaborative activities, this paper presents a developed co-design approach for early engagement in exploring the key features and characteristics that could be integrated into an MR application for assembly instructions. During two co-design workshops, the participants are engaged in reflecting on their work practices, defining their needs and desires, and visualising their assembly processes and craft skills. Using photo elicitation and tangible design methods as sensitising and generating sessions, the workshops aimed to understand the company's current practices, the areas which might benefit from implementing new technologies, as well as the training practices in the assembly processes. These insights provided the base for co-designing an early MR paper prototype, providing support for a later application on Microsoft HoloLens 2 assembly instructions. The findings also uncovered central, often neglected aspects in relation to the applicability and implementation of the new technologies in this context.

2. Theoretical perspective and related work

The co-design approach provides the conditions for people's participation, building on their experiences and providing the necessary tools for them to be able to act in their current setting (Bødker, 1996) and generate future alternatives (Sanders and Stappers, 2012). The generative methods, applied in this study, are one of the multiple categories of tools and techniques used in collaborative design practices. They are based on introducing tangible artefacts (toolkits) to support people's expression of both the emotional and the subjective experience, directly linked to our visual ways of expressing, knowing, sensing, and remembering (Sanders, 2000). In collaborative contexts, the tangible artefacts work as boundary objects, activated through making activities for collective exploring and testing hypotheses about future ways of living (Brandt et al., 2012). Applying these artefacts in relation to emerging technologies, the objects provide support for the people who lack computational expertise in exploring and imagining new smart products of the future (Bozic-Yams & Aranda-Munoz, 2021).

The MR technology has been emerging in recent years, gaining more significance in different fields and changing work practices (Park et al., 2021). Combining the physical environment with computer-generated information, MR technology allows the users to interact with virtual objects in the real world (Qamar et al., 2023). Studies following a collaborative approach for designing MR applications have been applied in several contexts, such as healthcare, education, architecture, manufacturing industry. Often the collaborative approach is implemented by engaging potential users or stakeholders in interacting and testing the MR simulations designed beforehand, providing verbal or written feedback from their immersive experiences (Dastan et al., 2024; Maurya et al., 2019). By that, the participants are able to directly influence the design and support its future development. However, limited research is focused on providing tools for engaging the potential users in the early design phases of MR applications, before the digital simulation is established (Curtis et al., 2024; Venemyr, 2024), particularly in manufacturing industry contexts. With this paper, we aim to place the participants in the centre of the development and application of MR technology. We explore the co-design approach for early engagement through tangible methods, by providing space for the participants in reflecting, visualising and defining their own ideas and experiences in relation to future technological implementations.

3. Empirical study

3.1. Background

This study presents the co-design approach within a small manufacturing company, supported by a technology partner, as part of a research consortium funded by the European Commission (EU), that aims to analyse the human-machine relations, defining future pathways needed to succeed in this space. The study was conducted over a period of one year, between 2023-2024, during which several meetings, co-design activities and testing sessions occurred. The company's production and business model have developed over the years from a craft tradition including fine mechanics, blacksmithing and lock manufacturing. The workforce today consists of twelve people in addition to administrative staff, where five senior operators have more than thirty years of experience in the lock industry. With

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a “low volume – high mix” business model, an important issue is the need to maintain skills and competences that are relevant to the manufacture and assembly of products, but also professional knowledge of how old locks are produced and assembled. Their aim is to improve the inadequate, and for some products non-existent, instructions on how to effectively assemble mechanical lock components.

The management expected that an MR-based methodology would support the flexibility requirements of the business model, increasing rotation of assemblers between different product series and workstations, and the attractiveness of their industrial craft skills and motivate younger potential employees. The technology partner proposed a pilot test for implementing the MR technology as an alternative to their current lock assembly instructions (Köping Olsson, 2024). During the initial training session, one senior operator, with over thirty years of experience, tested the Microsoft HoloLens 2 tool at the technology partner’s headquarters. He first assembled a test product following ready-made instructions, and then experimented with designing his own assembly instructions for one of their locks. After this session, the company lost interest in the MR application, motivated as being inopportune for their company’s organisational and work practices. Thus, we decided to explore how a co-design approach will enable people’s engagement in reflecting on their assembly practices in relation to new technologies, and be an active part in designing the MR assembly instruction.

3.2. Methods

This paper focuses on two co-design workshops that provide valuable insights on the early participants’ engagement in exploring the potential implementation of new technologies, with focus on MR application. The workshops were organised on two distinct days, took place in a traditional conference room at the company factory and lasted for approximately two hours each. Both workshops followed the same structure, starting with a sensitising session, using the photo elicitation method and continuing with a generative session, applying the tangible method, for reflecting, visualising and defining different aspects of the assembly processes. The distinctive aims of the methods (Figure 1) are presented below under the respective subsection.

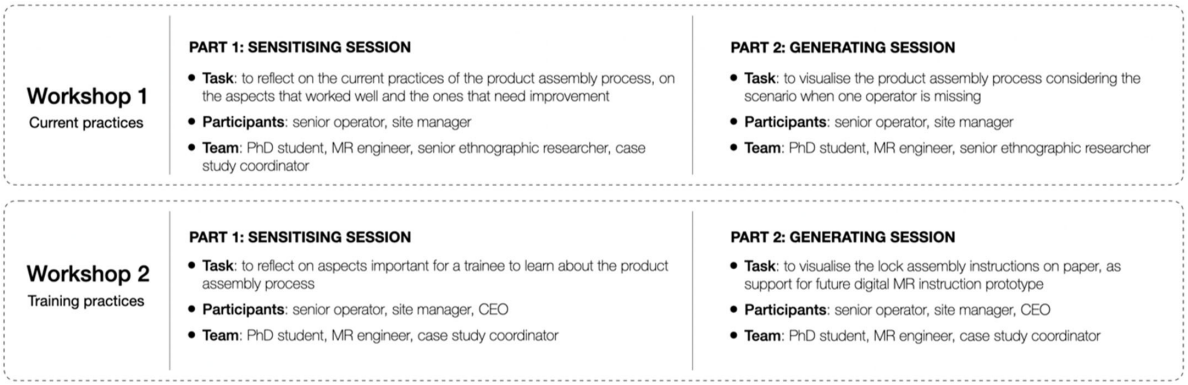


Figure 1. Workshops structure, task description and people engaged

The people participating in the workshops have also been part of the entire study, and consist of three representatives from the manufacturing company, having different roles: the senior operator, the site manager and the CEO. The organising team includes four people: the PhD student in Innovation and Design (designing and facilitating the workshops), the case study coordinator, the MR engineer and the senior ethnographic researcher (the last three observing and interacting in some parts during the process). Their presence during the workshops varies to some extent (Figure 1). The participants are seen as experts in their area, a view that has long been used in participatory design practices, with Sanders and Stappers (2012) recognising the people as the “experts of their experience”, having an active part throughout the design process, in developing knowledge, generating ideas and making design concepts.

The workshops’ data has been collected through audio recordings, photos, short videos, field notes and tangible artefacts.

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3.3. Sensitising sessions

The photo elicitation method is used during the first part of each workshop, as a sensitising session to immerse the participants in making observations and reflecting on their assembly processes, opportunities, challenges, that will be addressed later in the journey (Sanders & Stappers, 2012). This method has its roots in ethnographic research and as the participants take their own photos within their workspaces, it enables the possibility of communicating about their work practices through both verbal and pictorial, unfolding multimodal and multi-layered stories and lived experiences (Schaeffer & Carlsson, 2014).

During the workshops, the sensitising sessions follow the same structure. The participants are asked to take four photos in the factory, describe them using a template, and share their thoughts in an unstructured group interview. Designed and printed beforehand, the template contains two A4 pages of instructions: a short description of the task, two questions ("What did you take a photo of?", "Why did you choose this photo?") and a space for writing five keywords related to the photo (Figure 2, left). The session starts by introducing the task and the participants leave the room for approximately 20 minutes in search for the scenes, spaces or objects that would reflect their ideas, taking the photos using their own smartphones. Returning to the room, the participants write their reflections on the template for about 10 minutes, and share their photos with the PhD student via email (Figure 2, right). For the next 20 minutes the participants show their photos projected on the conference room screen, and express their thoughts in an unstructured group interview, encouraging open dialogue, comments and reflections.

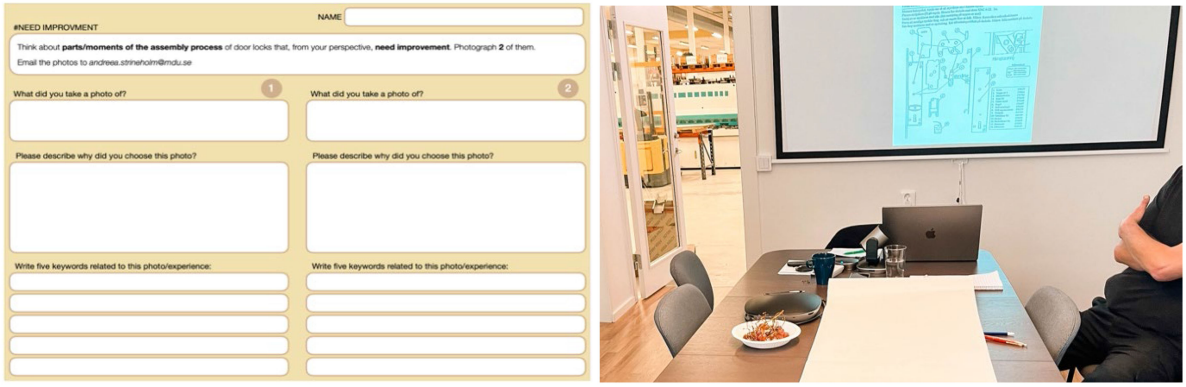


Figure 2. The photo elicitation template (left). Sharing moment during the session (right)

The first workshop centres around reflecting on the current practices of their product assembly processes, the participants being asked to think about the parts and moments of the assembly process that, from their perspective, worked well, and the ones that needed improvement. The second workshop narrows the assembly process toward their training practices, taking four photos related to the most important aspects a new employee should learn during the training process.

3.4. Generative sessions

During the second part of each workshop, tangible methods were used to support the participants' engagement in exploring and generating alternatives in their everyday work processes through making. Tangible methods enhance the participants in communicating their ideas and visualising their thoughts that otherwise would be difficult to articulate verbally, creating also a shared language between the participants and the team (Sanders and Stappers, 2012). Interacting with the materials and creating tangible artefacts also provides support for constructing meaning and new ways of understanding (Tversky, 2011), as well as collecting and preserving information (Mäkelä, 2005) in relation to the participants' assembly practices, craft skills and detailed procedures. To facilitate the engagement, two toolkits are made prior to the workshops. This method starts by introducing the task to the group, inviting the participants to use the toolkit arranged on the table (Figure 3, left), to visualise their ideas, processes and visions. The making part takes around one hour, supporting shared discussions, reflections and questions within the group, regarding the topic (Figure 3, right). The method ends with a 10-minute group reflection and agreement for the next activity.

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Figure 3. The toolkit from the workshop 2 (left). Using the toolkit during the session (right)

The first workshop relies on understanding their current assembly process in relation to one specific situation. The participants are asked to imagine the scenario in which one of the operators is missing for a few days, and someone else has to produce the orders. To visualise this process, the toolkit contains multiple elements used as triggers that allow room for a variety of interpretations, including simple and symbolical shapes (wooden blocks and sticks, wooden figures, face figures, coloured round stickers), writing tools (coloured markers and pens), paper tools (post-its, paper tape) and one long white paper for the base. When the complete process is visualised, the participants are invited to reflect and mark on the paper, the parts that would benefit from the implementation of new technologies in general, and then MR technology in particular.

For the second workshop, the participants engage in visualising the assembly instructions of one lock, marking all the steps and particularities that someone unfamiliar with the process would need to complete the task. To make the instructions, the toolkit includes specific and symbolic shapes, designed in Adobe Photoshop and Keynote, printed on paper and individually cut (lock components from the original instructions; symbolic shapes of photo and video cameras, headphones, microphones, speakers, magnifying glasses, arrows, check marks, stops, hearts, stars; plain rectangles and circles). Other elements complete the paper shapes, such as foam face figures, paper tools (post-its, paper tape and scissors), writing tools (coloured markers and pens) and one long white paper for the base. These materials are selected to both resemble the physical lock (paper lock components), but also relate to the future digital experience (interaction and emotional symbols), in order to visualise each assembly step on paper, providing further reference when designing the Microsoft HoloLens 2 instructions. The toolkit also contains the original lock components and tools to be assembled in parallel to prototyping the steps on the paper.

3.5. Analysis

For the analysis, the audio recordings of both workshops have been transcribed using Amberscript, manually corrected and translated from Swedish to English. Along with the transcripts (a total of 95 pages), we also looked at the visual material generated during the workshops, such as the photos taken by the participants during the sensitising sessions (a total of 20 photos), the fill-in templates (a total of 10 pages), the artefacts produced during the generative sessions (one for each workshop), and the photos and short videos taken by the PhD student. With various data, we present the findings following an inductive approach, describing the participants' engagement in the early phases of co-designing the MR assembly instruction. We look at the ideas regarding their current practices and training practices they engaged with during each workshop.

In the discussion session, we reflect on how the co-design approach provided support for the participants' early engagement in relation to both the consortium and the company aims, using tangible and digital materials for reflecting, visualising and defining their ideas and practices.

4. Findings

4.1. Engaging in current practices

During the first workshop, the sensitising session uncovers several aspects of the participants' current work practices within the lock company. Despite the task being narrowed towards the assembly processes, both participants choose to expand the reflections on the production level. For

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example, when reflecting on what needs improvement, the senior operator lifts his struggle regarding their constant process of improving the assembly technical details, which is dependent on an external provider who owns the components. Another example is brought up by the site manager, in relation to a recently implemented production support system. He thinks this new production support system "has isolated" them, as the information displayed (on a computer screen) is too narrow, being difficult for the operators to make the right decisions when choosing an order and requiring more experience in navigating the system. When reflecting on the aspects that work well, the participants mention the good collaboration among the employees, their professionalism and willingness to continuously develop their competences and skills, as well as their working spaces that support their qualitative work.

The generative session also provides insights on their current process, which oriented the design of the MR assembly instructions, along with uncovering another technological need. The participants used the toolkit to reflect on the occasions the senior operator might need assembly instructions, visualising a specific moment when one of their colleagues was sick and he had to cover his orders and assemble products that he was not familiar with. As during the sensitising session, instead of only focusing on the assembly processes, the participants use the materials to show the entire process of completing this order, from choosing it to making it ready for the delivery, by moving the physical elements, enacting the wooden figurines and writing down their routines on the large paper (Figure 4).

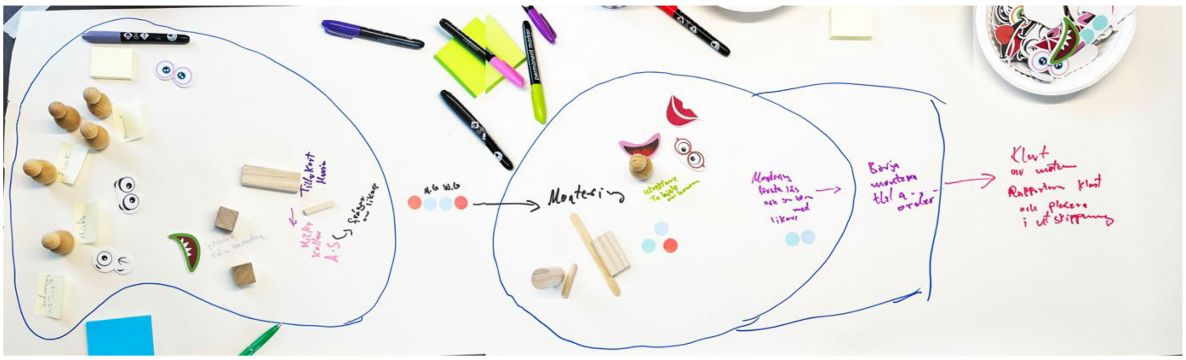


Figure 4. Current production process and technologies aspects visualised by the participants

Throughout the session, the participants explain each of the steps the senior operator makes in order to assemble the order, such as looking for the instructions if they are available (the site manager explains that today, only 40% of their products have assembly instructions or standards), checking the ready assembled product used as a reference for that specific lock, consulting colleagues who are more familiar with these specific components, and based on all this information, applying his own assembly procedures. The participants bring up their engagement in improving their instructions by constantly checking all the parts to find more effective processes. This is both an individual and a collective activity, challenging the employees in developing their skills and innovative processes.

To better understand their needs, the participants are asked to reflect and mark on the paper the areas where a possible new technology might support their current production processes. They highlight the need of a customised production support system that would be interactive, provide clear information and support the operators in understanding the planning and making active decisions. When asked about the possibility of MR instructions, the senior operator acknowledges that for him written instructions are more useful, instead of using the Microsoft HoloLens 2 tool, which the site manager suggests might provide support for younger trainees, with no previous experience, in their apprenticeship processes. They also discuss the possibility of a paper or a video format more relevant for the senior operator, with key features such as being available without asking someone else, easy to navigate or skip the steps, and including fine details important to remember. We conclude by highlighting that they need two distinctive types of instructions, one for the trainee (with possible MR application) and one for the experienced operators (to quickly remember the process).

4.2. Engaging in training practices

The second workshop starts with the team presenting a summary of the insights gathered in the previous workshop, as the company CEO joins the group. We show the two production areas with potential for

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new technology implementation: the production support system and the instructions, with their two distinctive applications for trainee and experienced operators. We discuss extending the current collaborative study towards the production support system, and the participants share their interests and initial ideas regarding this possible technological implementation. Following that, we mutually agreed on continuing the exploration of co-designing the MR assembly instructions, with focus on training.

The sensitising session brings insights on the main aspects in training a new employee, the participants reflecting on their organisational values, training practices and production routines. For example, all participants highlight the need for a new employee to join the group and become a team member. The CEO considers the social criteria as a crucial aspect for getting employed. They also value the willingness to learn, support and respect each other in their dynamic team. Learning the correct instructions and procedures is highlighted from different views. The site manager describes their training as a "side by side" process with the experienced trainer, while the senior operator mentions the importance of individual discoveries in learning how to navigate the space. Other aspects such as safety and qualitative requirements and protocols are also important. These insights provide the framework for the following prototyping session.

The generative session engages the participants in prototyping the assembly instructions of one of their locks, by observing, reflecting and mapping all the necessary steps. (Figure 5). The toolkit is intended to be used by the participants, but after introducing the task, none of them is willing to pick up the paper components and make the instructions collage. Therefore, the participants and the team members start a collective prototyping process: the senior operator shows his assembly process on the original lock, the PhD student maps the process using the toolkit, and everyone gets involved in sharing knowledge, asking questions and reflecting on the process, directly linked to their own expertise.

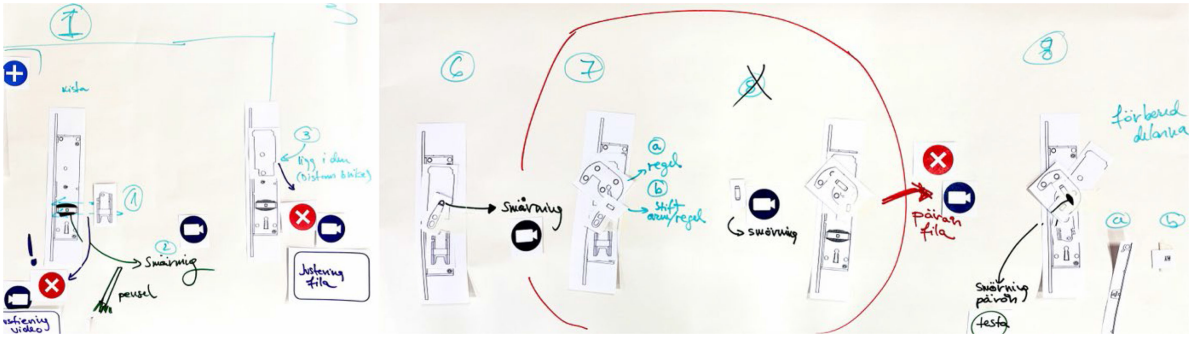


Figure 5. Assembly instructions: paper prototype

Thus, our prototyping process includes the following actions: we observe the apparent and subtle movements of assembling the original lock and reflect on them; we tape the paper components on the long paper, following the same order as in the original lock assembly; we ask clarifying questions, as the original lock assembly movements might seem obvious or routinised for the experienced ones; we agree on including all the small details that might appear during the process, and decide what to keep and what to leave out later, when making the Microsoft HoloLens 2 instructions; we number the steps and constantly reflect on their length and details that should be included in each one; we identify the moments when particular problems or exceptions might occur, and mark the procedures to overcome them, as complementary information in the instruction process; we video record the original lock assembly movements, and mark the action on paper, under the respective step; we create a pattern on how to use the paper symbols, depending on the details that need to be represented; we reflect and negotiate how to connect the paper prototype process with the digital functions of the Microsoft HoloLens 2 tool, through different communication modalities (text, audio, video, 3D animation); we clarify the words and terminologies used by different people in the assembly process, and write the appropriate terms on the paper; we look for patterns and particularities in mapping out the steps; in the end, we review all the steps, micro procedures and complementary information, to ensure its correctness.

We finish the generating session by displaying the three-metre-long paper prototype outside the conference room, in the factory space and have a short reflection moment with all the participants and team members on the process, planning the next activities and checking the additional information that might be needed in creating the Microsoft HoloLens 2 instructions. We also took a complete video of the senior operator assembling the lock, at his regular workstation.

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5. Discussions and future work

During the workshops, the co-design approach provided support for engaging the participants in reflecting, visualising and defining their work practices, needs, desires, and craft skills. As this study was conducted within the framework of a larger consortium, the following model (Figure 6) shows the consortium and the company aims emerging through the co-design approach, enabling the exploration of situated technology applications.

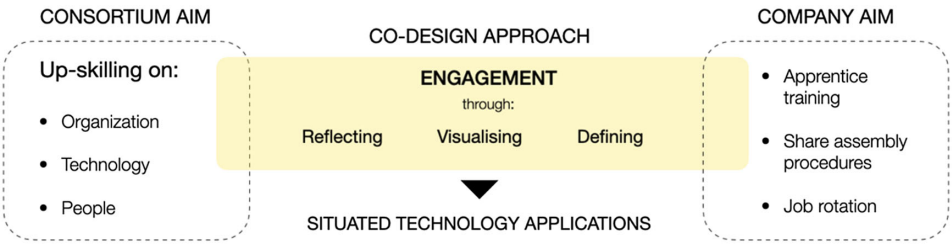


Figure 6. Engagement model through co-design approach

Rather than starting the MR implementation by following the initial plan between the consortium representatives and the company management (the operators to learn and apply the Microsoft HoloLens 2 instructions designed by the technology partner), the co-design approach provided the base for bringing in a more systemic perspective within the manufacturing industry, where the technology and organisation are best designed in an integrative way with consideration of people's needs, abilities and limitations (Rösiö et al., 2021). Reflecting, visualising and defining were the key engagement practices that emerged throughout the co-design workshops. Reflecting, visualising and defining through both tangible (the toolkits used during the generating sessions) and digital (the photos taken by the participants during the sensitising sessions) materials, encouraged the participants to engage through different ways of communication, forming new meanings, giving voices to their artefacts and creating a shared understanding (Brandt et al., 2012; Mäkelä, 2005; Schaeffer & Carlsson, 2014) of their production and assembly processes. Besides the expected result in creating a physical prototype for the MR assembly instructions, we also uncovered other aspects of the participants' production practices that complement their assembly practices, which might have a higher influence in improving their work.

The workshops structure also played an important role in how the participants engaged during the activities. Using the photo elicitation method during the sensitising sessions immersed the participants in making observations of their current and past experiences, that were later used as a base for bringing up the future perspective (Sanders & Stappers, 2012). Using their own mobile phones to take photos of their own environment encouraged the participants to start the workshops with a more familiar way of conducting the activities. During the following generating sessions, the insights uncovered during the sensitising sessions became part in the making of the tangible artefacts, as for example during the first workshop when the issue of the production support system was marked on the paper as the primary need for technological development within the participants' process. During the second workshop, the idea of learning the correct procedures was key in marking the steps on the MR paper prototype.

Applying the co-design approach engaging with digital and tangible materials also provides flexibility and space for the participants to use the methods in a way that encourages their own forms of communication (Sanders & Stappers, 2012). An example here is when one participant adapts the photo elicitation method to make it comfortable for his own way of communicating, passing over the part on writing reflections on the template, and verbally sharing his reflections in connection to the photos. Another example is observed during the first workshop, when despite having a clear task to support framing their ideas around the assembly process, the participants chose to expand the task and discuss about their larger order production processes. Taking a broader approach, we understood that we cannot isolate the assembly process (and instructions), as they are a small, interconnected part within a larger production process. A third example took place during the prototyping session of the second workshop, when the toolkit was intended for the participants to use by themselves for mapping the instructions steps, but it became a more collective session, with the facilitator placing the materials on the paper, the senior operator assembling the physical lock. The other participants and team members were joining at certain moments, sharing their own perspectives, expertise and understanding of the micro procedures, negotiating symbols, formats and vocabulary, creating a common view on the final paper prototype.

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By early engaging in co-designing the MR assembly instructions using tangible and digital materials, the participants could reflect, visualise and define the company's practices, needs, opportunities and key features in implementing new technologies (Jin & Fagan, 2024), uncovering the relations between their assembly procedures (and instructions) and production process, their team and work environment, and the new technological support and their different users. The co-design approach identified the premises for designing the MR application of the assembly instructions prototype with the apprentice training in mind, as well as bringing up the possibility for expanding the case study framework towards the customised production support system.

As this study focuses on the co-design approach in the early phases of MR application, uncovering central aspects of the company perspectives on new technological applications, we suggest as future research to further collaborate in designing, developing and testing the Microsoft HoloLens 2 assembly instructions in a training setting, and how it is influencing the learning outcome, based on both the specificity of the procedures, and the continuous development of the operators' competences and skills. Evaluating the implementation of the MR technology could be conducted by applying a more longitudinal ethnographic approach. We also plan to further develop the co-design approach for engaging the participants in developing the production support system brought up during the workshops, developing other sets of materials and toolkits that would adapt to the new technology. Future research is further needed in order to explore how the methods presented in this paper could be applied in other contexts, to engage the participants in the early phases of the MR implementation. How to design the toolkits to relate to the digital MR application but still allow space for flexibility, how this approach will influence the final design, what additional aspects are lifted up during the process and how to navigate them in a broader research consortium.

6. Conclusion

This study provided key insights into the diverse challenges and needs that can arise when involving the people in early exploration of new technological development, emphasising the need for the human-centricity design approach, one of the core principles of Industry 5.0. Following a co-design approach implemented through tangible design methods and photo elicitation method, we provided space for the participants for reflecting, visualising and defining aspects of their current practices and future ways of working. The tangible artefacts worked as boundary objects that allowed people to engage in the early co-design process. We learned that while MR gains popularity, it might not be the most suitable solution of assembly instructions for the experienced operators, but it might possibly provide support in the apprenticeship training process. We shifted then towards co-designing an MR assembly instructions paper prototyping for later Microsoft HoloLens 2 application, by reflecting, defining and visualising the necessary steps and detailed procedures for a new trainee in learning to assemble the lock. We also uncovered other areas of technological implementation, setting the framework for co-designing a customised production support system. This highlights the importance of following a co-design approach, enabling dialogue and increasing awareness of work practices, supporting the participants in identifying situated areas for new technology applications and co-designing the solution development.

Acknowledgements

The work described in this publication was subsidised by Horizon Europe (HORIZON) framework through HORIZON-CL4-2021-HUMAN-01 call - A HUMAN-CENTRED AND ETHICAL DEVELOPMENT OF DIGITAL AND INDUSTRIAL TECHNOLOGIES 2021

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https://doi.org/10.1017/pds.2025.10155 Published online by Cambridge University Press